6.1 Derivative function and standard functions

Definition and results

A function is differentiable on an interval when it is differentiable at each of its points; its derivative assigns to x the number $f\prime(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)\prime = nx^{n-1}$ on its domain of differentiability; for n = 0, the derivative is 0 directly.

Status and justification. Square: the rate is 2 $a + h$. Reciprocal: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are assumed here.

Worked example. For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f\prime(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function is defined at 0, but its usual derivative exists only for $x > 0$.

Watch out

The set on which a function is differentiable may be smaller than its domain of definition.

Immediate application. Diff. x^5, x^{-2} & $\sqrt{x}$ stating the domains.

6.2 Sum, product, reciprocal and quotient

Definition and results

For differentiable u and v, ($u + v)\prime = u\prime + v\prime$, $(\lambda u)\prime = \lambda u\prime$, $(uv)\prime = u\prime v + uv\prime$. Where $v \neq 0$, $(\frac{1}{v})\prime = -\frac{v\prime}{v^2}$ and $(\frac{u}{v})\prime = \frac{u\prime v - uv\prime}{v^2}$.

Status and justification. For the product, add and subtract $u(a + h)v(a)$ in the numerator. The rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. Since a differentiable function is continuous (assumed), we obtain $u(a)v\prime(a) + v(a)u\prime(a)$. The other rules are accepted without proof.

Worked example. For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f\prime(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping the brackets avoids changing only one sign instead of the whole numerator.

Watch out

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Immediate application. Differentiate $(x^2+1)(3x-2)$ then $\dfrac{1}{x^2+1}$.

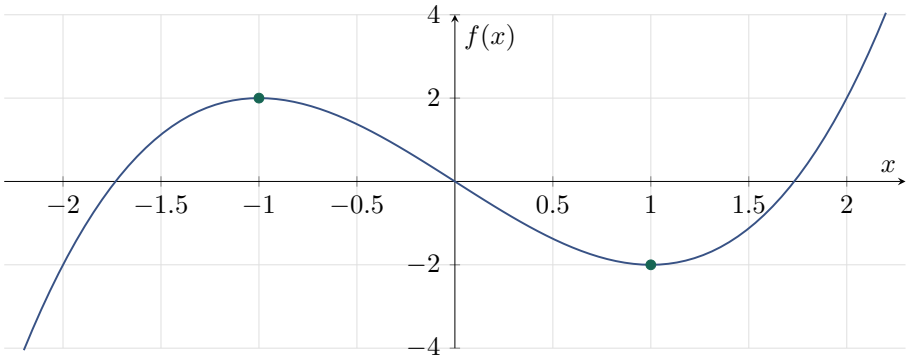
6.3 Sign of the derivative and extrema

Definition and results

On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies that f is strictly increasing. If f has a local extremum at an interior point where it is differentiable, its derivative there is zero. The converse is false.

Status and justification. Variation theorems assumed. For an interior minimum at a, the rates on the right are positive or zero, those on the left negative or zero; if they converge to the same real number, it is 0. Same argument for a maximum.

Worked example. $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $]-\infty; -1]$, decreases on $[-1; 1]$, increases on $[1; +\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. None of these extrema is global on $\mathbb{R}$.



$f(x) = x^3 - 3x$: a local maximum at -1 and a local minimum at 1.

Watch out

The theorem applies on an interval. A function may be decreasing on two separate intervals without decreasing on their union.

Immediate application. Study $f(x) = x^2 - 4x + 1$ on $[0; 3]$ and give its extrema on this interval.

6.4 Parity, inequalities and comparison

Definition and results

On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The graph of an even function is symmetric about the y-axis; that of an odd function about the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

Status and justification. The symmetries follow from the coordinates of the image points. For an inequality, a study of variations can give a minimum of the difference, then its sign; the derivative does not directly replace this difference.

Worked example. For $x > 0$, $h(x) = x + \frac{1}{x} - 2$ has derivative $\frac{x^2 - 1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, so $x + \frac{1}{x} \geq 2$. Equality holds only at 1.

Watch out

Check the domain before parity. The sign of $f' - g'$ gives information about the variation of $f - g$, not directly about its sign.

Immediate application. Study the parity of $x^4 + x^2$ & $x^3 - x$; prove $x^2 + 1 \geq 2x$.

6.5 Optimisation and Newton's method

Definition and results

Optimising consists of defining the variable and its domain, expressing the quantity to be optimised, then comparing all the candidate values, including the endpoints. Newton's method uses the tangent to approximate a solution of $f(x) = 0 : x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ when $f'(x_n) \neq 0$. It is used only in favourable cases.

Status and justification. Newton's formula comes from the x-intercept of $y = f(x_n) + f'(x_n)(x - x_n)$. Convergence is not guaranteed in general; it is observed and checked in the case considered.

Worked example. For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton's method gives $x_1 = \frac{3}{2}$, $x_2 = \frac{17}{12}$. The squares are 2.25 then about 2.00694: we approach $\sqrt{2}$. The program must check for zero derivative and max number of steps.

Watch out

A numerical value given by an algorithm is not proof of a minimum or of convergence.

Immediate application. A rectangle with perimeter 12 m has width x. Find the maximu
by differentiation.